# Voltage Reference and Zero Current Detector Monolithically Integrated on p-GaN Technology Designed for Process Corners Compensation

Plinio Bau*, Sebastian Gavira-Duque*, Frederic Rothan†, Cedric Reymond* and Dominique Bergogne*
Email: plinio.bau@wise-integration.com
*Wise-Integration, 38240, Meylan, France
†Univ. Grenoble Alpes, CEA, LETI, F-38000 Grenoble, France

***Abstract* — This work describes two designed and fabricated circuits for power device integration in 650 V e-mode GaN-on-Si technology and gives their main measured characteristics. The first circuit is a voltage reference designed to compensate process, voltage and temperature (PVT). The advantage of this circuit over state-of-the-art circuits is its performance of regulation, low current consumption and surface required (only 100 µm × 60 µm in the first prototype built). The second circuit is a zero-crossing current detector that is also designed to be insensitive to process variations. It exhibits stable static characteristics and bandwidth observed in simulation of more than 10 MHz. The convenience of this circuit is its simplicity because no differential amplifier is used to make a comparison. Both circuits are designed to be integrated with power transistors in GaN technology to be used in ACF or other topologies capable of switching at 650 V.**

***Keywords—GaN Power Transistors, p-Gate AlGaN/GaN HEMTs, Voltage Reference, Current Detector, Design for Manufacturability.***

## I. Introduction

Voltage reference circuits in monolithically integrated circuits in GaN-on-Si with p-type GaN AlGaN/GaN HEMTs (normally-off) [1-2] present the challenge of being very sensitive to process variations and charge trapping effects. To design circuits in GaN technology with performance and stability similar to CMOS [3-4], designs must compensate process variations that affect mainly the threshold voltage Vth and the mobility [5-9]. Another challenge is that in this technology there is no transistor equivalent to PMOS. Gate driver integration in the GaN-on-Si technology [10-12] needs analog circuit blocks to provide circuits such as UVLO, LDO, and input compatibility circuits. Current detectors can be helpful for short circuit detection and ZCS switching converters.

## II. Voltage Reference Circuit

### A. Schematic

The circuit presented in this work (Fig. 1), generates a stable output voltage around 3 V in typical conditions. The output voltage is nearly insensitive to the supply voltage, temperature and process variations. A further advantageous feature of this circuit is low sink current value measured at 3.71 µA. The other advantage is that the voltage reference can drive loads of impedance as low as 10 kΩ with a variation of only 6% in the output voltage ($V_{REF}$), therefore this circuit can be easily adapted to make LDO circuits. If more current is needed, the first depletion mode transistor (M1 in Fig. 2) can supply them by changing slightly the gate-source voltage without impacting the voltage in the rest of the circuit.

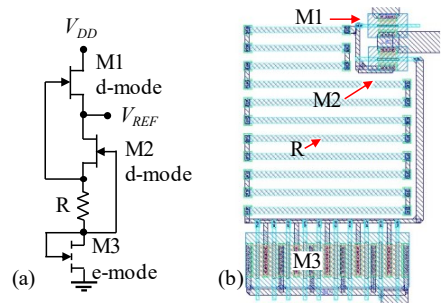


Fig. 1. Voltage Reference (a) schematic and (b) layout. The total area implemented in GaN-on-Si 0.5 µm technology is 100 µm × 60 µm.

### B. Comparison with the state-of-the-art

The table I below compare the parameters of the voltage reference to the state-of-the-art voltage reference [1]. The area required to build this circuit not using dummy structures is 86 times smaller than the state-of-the-art. This circuit also presents a good behavior for process corners variation while de state-of-the art circuit didn't report any study regarding this subject.

TABLE I. Comparison between new voltage reference to the state-of-the-art.

| Parameter | ISSCC 2020 [1] | New Volt. Ref. |
|---|---|---|
| $V_{REF}$ | 3.19 V | 2.81 V |
| $V_{DD}$ range | 3.9-24 V | 3.1-20 V |
| Line sensitivity | 0.32 %/V | 2.2 %/V |
| Temperature coef. | 50-200 ppm/°C | 256 ppm/°C |
| Process variation | N/A | 4.4% |
| Load variation | N/A | 7.71% |
| Current consumption | 3.1 µA | 3.71 µA |
| Area | 0.52 mm² | 0.006 mm² |

### C. Design and simulation of the P-GaN voltage reference circuit

When the current flowing through a depletion mode transistor with a resistor connected between gate and source, is much lower than the saturation current, the gate-source voltage has a tendency to remain constant as the threshold voltage ($V_{TH,D}$). Similarly for a transistor connected instead of the resistor. The configuration presented in Fig. 2 (a) works as a current source connector do a resistor R2 and so then the voltage at the top node of this resistor is constant whatever is the voltage applied to M1 drain ($V_{DD}$ node).

2023 IEEE Applied Power Electronics Conference and Exposition (APEC) | 978-1-6654-7539-6/23/$31.00 ©2023 IEEE | DOI: 10.1109/APEC43580.2023.10131398



Authorized licensed use limited to: Tsinghua University. Downloaded on February 13,2026 at 02:45:44 UTC from IEEE Xplore. Restrictions apply.

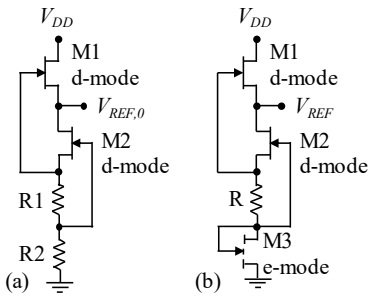


Fig. 2. (a) Constant current source with normally-on transistors. (b) reducing the fabrication dispersion by adding a normally-on transistor (M3).

The overall idea is to obtain a constant drain-source voltage of M1. Circuit of Fig. 2 (a) provides a constant current in R2 and a constant voltage, however when process dispersion is taken into account (fabrication corners FF, SF, FS, SS), the output voltage value $V_{REF,0}$ is highly impacted. The threshold voltage ($V_{TH,D}$) can have variation of $V_{REF,0}$ by ±50%, see Fig. 4 (a). It is important to limit the current in the depletion mode transistors to avoid strong variations in the voltage of the enhancement mode. By replacing R2 to one normally-off transistor M3, see Fig. 2 (b) and Fig. 4 (b), the $V_{REF}$ process dispersion is reduced to ±4.4%. Furthermore, the threshold value of both transistors (e-mode and d-mode) increases from corner FF to the corner SS. However, the magnitude of the d-mode threshold voltage value is always negative, which makes the absolute value of the magnitude to be opposite to the e-mode threshold value. See Fig. 3.

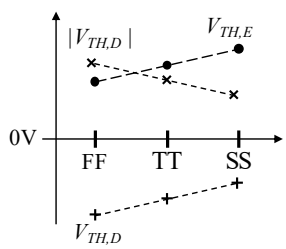


Fig. 3. Threshold voltage tendencies for normally-off ($V_{TH,E}$) and for normally-on ($V_{TH,D}$) value for different fabrication corners.

TT is the typical value; FF is one extreme deviation and SS the opposite extreme deviation. The $V_{REF}$ value is obtained by the expression (1) below. It means that for one corner the $V_{TH}$ is maximum for one and minimum for the other and vice versa in the other corner. In that way, piling up one e-mode and two d-mode GaN transistor the $V_{TH}$ corner's variation is compensated.

$$V_{REF} = V_{TH,E} + 2 \times V_{TH,D} \quad (1)$$

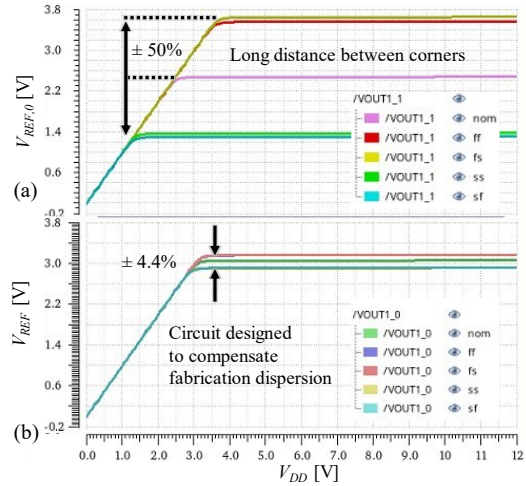


Fig. 4. Simulated $V_{REF}$ as a function of $V_{DD}$ for a circuit (a) before and (b) after compensated for corner's dispersion.

## D. Measured Performance

The Fig. 5 below is how to measure the input-output static characteristics ($V_{REF}$ vs $V_{DD}$) and using a script in python programming language to process the oscilloscope data and generate a resulting input-output waveform for each prototype chip. Fig. 6 shows the measurement of 9 different chips in the same plot.

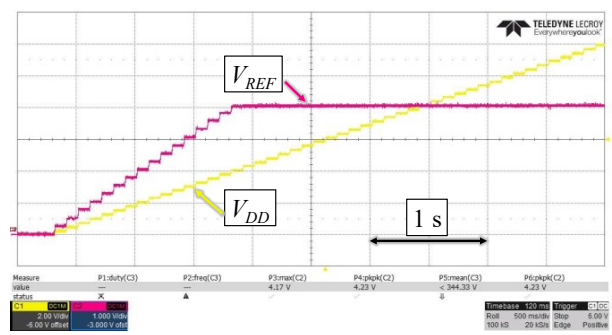


Fig. 5. Measured input-output characteristics by applying a quasi-DC (5 Hz) triangular wave.

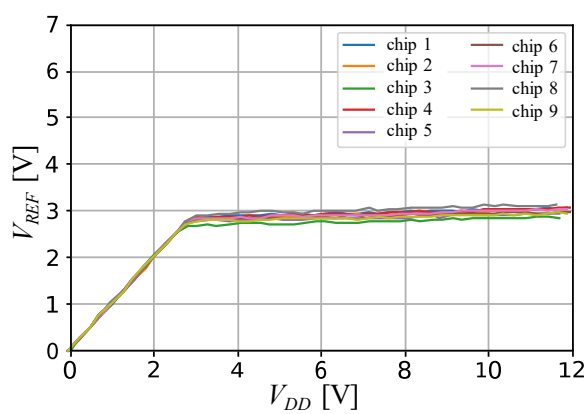


Fig. 6. Measured $V_{REF}$ versus $V_{DD}$ characteristics for 9 different fabricant prototypes located far apart in a same multi project wafer.



Authorized licensed use limited to: Tsinghua University. Downloaded on February 12,2026 at 02:45:44 UTC from IEEE Xplore. Restrictions apply.

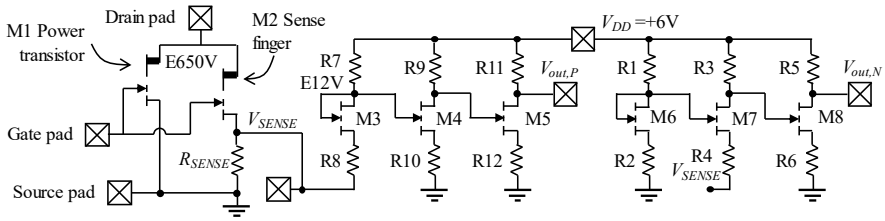


Fig. 7. Current detector schematic. A prototype is fabricated in 650V E-mode GaN-on-Si technology.

The table II below compare the simulation and measured parameters of the voltage reference. The measured values are the average of the chips with a behavior considered acceptable. The behavior is considered acceptable if the value of $I_{DD}$ is within the corner's simulation value. 2 chips out of 14 had the $I_{DD}$ supply current above 100 µA instead of the expected 3 µA. 11 out of 14 chips were within the range of pass and therefore the multi project wafer yield is 12/14 = 85%. Regarding temperature variation, two prototypes are placed in the oven up to 150°C and one shows 3.86% of variation from 25°C to 150°C and another one 2.55%. The average value is presented in the Table II. All other parameters are the average values of 11 chips. One load resistor $R_L$ is connected between the node $V_{REF}$ with the ground node.

TABLE II. COMPARISON BETWEEN POST-LAYOUT SIMULATION WITH MEASURED PARAMETERS.

| Parameter | Conditions | Post-Lay. Simulation | Measured average |
|---|---|---|---|
| $V_{REF}$ | $R_L$=open, $V_{DD}$=6 V | 2.924 V | 3.033 V |
| $V_{REF}$ | $R_L$=100 kΩ, $V_{DD}$=6 V | 2.813 V | 2.90 V |
| **Line Reg.** | $R_L$=open | 0.14% | 2.2% |
| **Line Reg.** | $R_L$=100 kΩ | 0.18% | 2.3% |
| **Temp. Var.** | $R_L$=100kΩ, 25°C-150°C | 1.66% | 3.20% |
| **Load Reg.** | $R_L$=10 kΩ-10 MΩ | 6.82% | 7.51% |
| **Current Cons.** | $R_L$=100kΩ, 25°C-150°C | 4.39 µA | 3.60 µA |

## III. CURRENT DETECTOR

Another circuit presented in this work is a zero-current detector circuit that is also designed for process variation compensation and can work in frequencies observed in simulation up to 10 MHz while switching 650 V. The output $V_{out,P}$ goes from 0 to 6 V (low to high logic) when the current rising edge crosses the near-zero threshold and similarly the current output $V_{out,N}$ goes from 0 to 6V when the current falling edge crosses the near-zero threshold. The threshold values are designed to be as close as possible to zero for all PVT conditions. The outputs are non-overlapping, and therefore can give redundant information for critical applications when connected to a microcontroller. The schematic of this circuit is presented in Fig. 7.

### *A. Design and simulation of the P-GaN voltage reference circuit*

The HEMT transistors M1 is a 650 V and 120 mΩ transistor including the bonding wire resistance. The sense finger transistor M2 is also rated to 650 V with 1 mm width. This sense finger transistor (M2) is placed in parallel with the main power transistor with only its source not connected together. One resistor using the R2DEG (2D electron gas resistor, very specific to GaN HEMT technology) fabrication technology with value 2.3 kΩ is used (and named $R_{SENSE}$) to generate a voltage waveform directly proportional to the drain-to-source current in the power transistor M1. The resistance value of $R_{SENSE}$ is considerably higher than the sense finger on resistance ($R_{dsON,M2}$). All resistors were implemented using the same R2DEG technology. All other transistors (from M3 to M8) are low voltage transistors and have a recommended maximum drain-to-source voltage of 12 V. However, characterization tests show these transistors can withstand up to 50 V before avalanche current is observed.

The first branch of the circuit is a common source amplifier with source degeneration. For sizing this branch, composed by resistors R1 and R2 biasing the transistor M6 the following expressions (2) and (3) are presented. By applying the Kirchhoff loop law, we obtain:

$$V_{DS,3} = V_{DD} - (R_1 + R_2) \times I_{DS} \quad (2)$$

$$R_1 + R_2 = \frac{V_{DD} - V_{DS,3}}{I_{DS}} \quad (3)$$

The voltage across drain-to-source of M6 plus the voltage across the resistors R1 and R2 is equal to the supply voltage $V_{DD}$. To obtain the saturation voltage $V_{DS,SAT}$ below the $V_{DS,3}$ the sum of the biasing resistors R1+R2 is therefore:

$$R_1 + R_2 \leq \frac{V_{DD} - V_{DS,SAT}}{I_{DS}} \quad (3)$$

The saturation voltage ($V_{DS,SAT}$) depends on the corner and can vary +-50% in this GaN-on-Si technology. By taking the saturation voltage of the slow-slow (SS) corner, all cases are covered and the circuit operates correctly even with a high dispersion of corners during fabrication. After extensive simulation, the values that present a good trade-off between delay and power consumption are chosen to be R1= 8 kΩ and R2=180 Ω.

Fig. 8 is a transient simulation for a sinusoidal current applied between drain and source of the power device. Notice the sharp zero current detection observed when the outputs $V_{out,P}$ and $V_{out,N}$ switches from 0 to 6 V for different fabrication corners (TT, SS and FF).



Authorized licensed use limited to: Tsinghua University. Downloaded on February 13,2026 at 02:45:44 UTC from IEEE Xplore. Restrictions apply.

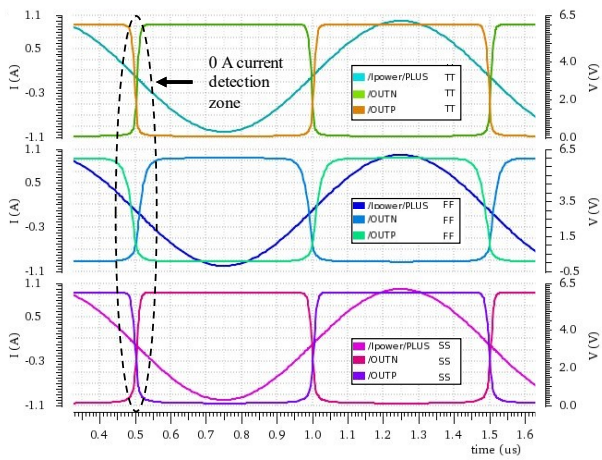


Fig. 8. Transient simulation of *VoutN* and *VoutP* as a function of *Iin* (the current in the Drain of the power transistor M1) for the 3 different corners (SS, TT, FF). The transitions are sharp with threshold value close to 0 V for all corners at 1 MHz.

In Fig. 9, a Monte Carlo simulation is performed to study the dispersion effect of the positive threshold value ($V_{out,P}$). This value is defined by the input node ($V_{SENSE}$) when the output crosses 3 V that corresponds to $V_{DD}/2$. The simulated average value for this threshold value is 26.63 mV and the standard deviation is 74.88 mV. Notice that designing with a different value of the biasing resistors R1 and R2 will change the threshold values and it could be calculated to be closer or more far from zero depending on the application.

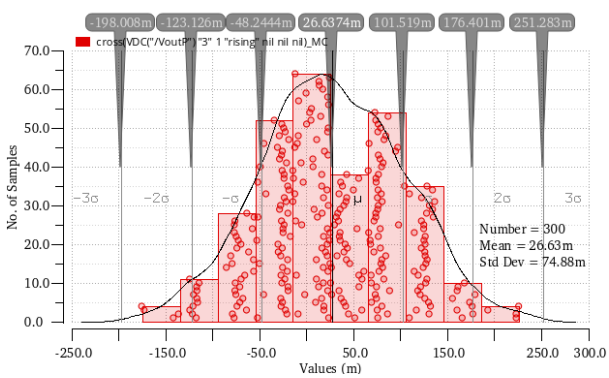


Fig. 9. Monte Carlo simulation for the threshold value (the input value of $V_{SENSE}$ when for $V_{outP} = V_{DD}/2$=3 V) for 300 samples.

## B. Measured Performance

Characterizations tests either impose a current in the drain pad of the 650 V transistor that generates a voltage on the $V_{SENSE}$ proportional to this current. Or the measurements are obtained by imposing directly a voltage value on the $V_{SENSE}$ test pad. In a final prototype, the $V_{SENSE}$ test pad does not need to be bounded to the package. For the first experimental test, a current is imposed using a source meter unit Keithley™ 2461 between drain-source of the power device and the $V_{SENSE}$ circuit node is measured and compared to simulated values for different fabrication corners (SS, FF and TT). The curve of $V_{SENSE}$ as a function of drain-to-source current in the 120 mΩ power transistor is presented in Fig. 10.

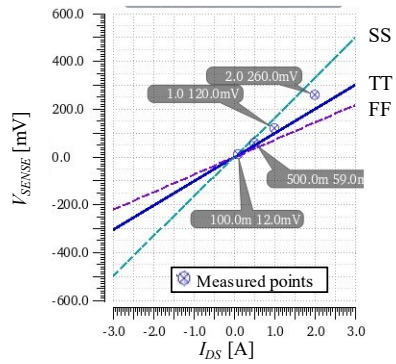


Fig. 10. Measured points and simulated curves for I-V characteristics of sense finger ($V_{SENSE}$) circuit node. The current is imposed on M1 with a source meter unit (SMU) and the voltage is measured.

The next characterization test involves imposing a constant value of voltage in the node $V_{SENSE}$ and observe the outputs switch from 0 to 6 V in a near-zero threshold value. Fig. 11 also correlates the expected simulated value with the measurements for 7 different chips located far apart in a multi-projects wafer. Fig. 12 is the equivalent for the negative detection output ($V_{outN}$).

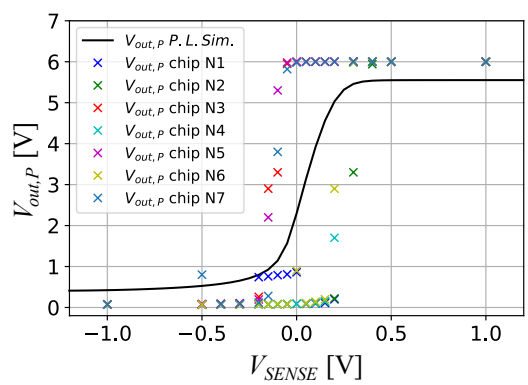


Fig. 11. Measured and post-layout simulated values for the positive current detector.

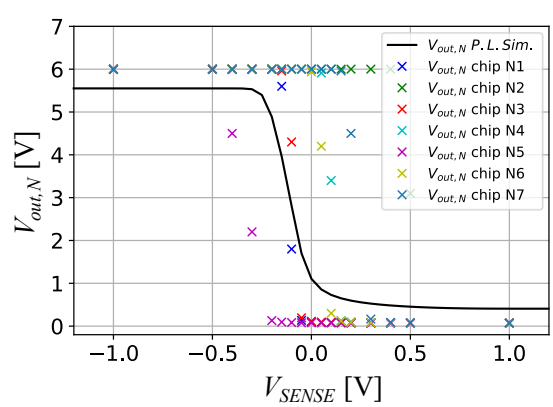


Fig. 12. Measured and post-layout simulated values for the negative current detector.

The measured data indicates the threshold value of the current detection is indeed close to zero. Some more



Authorized licensed use limited to: Tsinghua University. Downloaded on February 12,2026 at 02:45:44 UTC from IEEE Xplore. Restrictions apply.

optimization and layout improvements (common centroid and dummy structures) could improve further and reduce the dispersion of threshold value.

The advantage of this circuit for the corner's compensation is the pre-polarization of the biasing branch (composed by R1, R2 and M6) that places the operating point of M4 very close to the threshold value. For example, in the SS corner where the $V_{TH,E}$ of M6 is lower, the $V_{TH,E}$ of M6 will also be lower, therefore the threshold value of the circuit will be more incentive the process variation.

### C. *Current consumption of the current detector*

The first prototype is a proof of concept, the current consumption could be easily reduced, well below 1 milliamp. The power supply current reduction can be made by replacing the pull-up resistors to depletion mode transistors with a resistor connected in series with its source. A comparison of the current consumption with simulation vs measurements for 7 fabricated chips are plotted in Fig. 13.

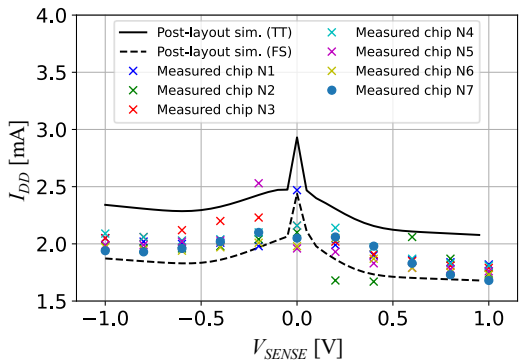


Fig. 13. Measured power supply consumption of the current detector compared to post-layout simulation values in the corners TT and FS.

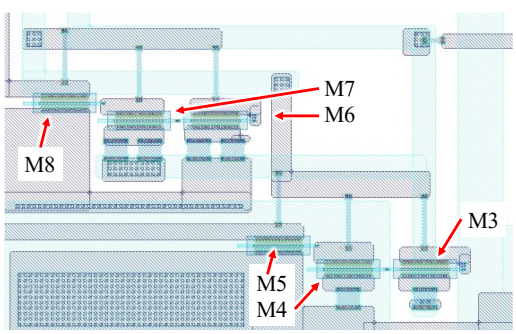


Fig. 14. Layout of the current detector (130 µm × 200 µm). The first prototype for proof of concept was not optimized in surface.

It is possible to observe the post-layout simulation for the typical fabrication corner (TT) and the fast-slow corner (FS) showing almost all measured data are inside this range. Fig. 14 shows the layout of the first protype that was also not yet optimized in surface.

## IV. Conclusion

A voltage reference monolithically integrated in GaN technology with considerably lower area comparing to the state-of-the-art, still having comparable PVT performance (process, voltage and temperature compensation) is presented in this paper. A current detector circuit that can be used in applications like zero voltage detection and zero current switching is also presented. Both circuits were fabricated and characterized. The measured results are compared to post-layout simulations. The contribution compared to the state-of-the-art is clear: smaller surface, less components required and fabrication dispersion study reported in this paper. These designs are simulated extensively for work in extreme fabrication corners (process variation). The characterization of main parameters is presented and all results correlate well with the simulations.

## References


[1] C. -H. Liao et al., "3.8 A 23.6ppm/°C Monolithically Integrated GaN Reference Voltage Design with Temperature Range from −50°C to 200°C and Supply Voltage Range from 3.9 to 24V," 2020 IEEE International Solid-State Circuits Conference - (ISSCC), 2020.

[2] Y. Ji, J. Lee, B. Kim, H. -J. Park and J. -Y. Sim, "18.8 A 192pW Hybrid Bandgap-Vth Reference with Process Dependence Compensated by a Dimension-Induced Side-Effect," 2019 IEEE International Solid- State Circuits Conference - (ISSCC), 2019.

[3] Y. Ji, C. Jeon, H. Son, B. Kim, H. -J. Park and J. -Y. Sim, "5.8 A 9.3nW all-in-one bandgap voltage and current reference circuit," 2017 IEEE International Solid-State Circuits Conference (ISSCC), 2017.

[4] J. -C. Du, Z. -G. Wang, X. Chen, J. Xu and B. -B. Ma, "Design of a Ring Oscillator with Temperature and Process Compensation Adopting a Novel Method," 2018 Progress in Electromagnetics Research Symposium (PIERS-Toyama), 2018.

[5] H. Çetinkaya, A. Zeki, A. Girgin and T. C. Karalar, "Composite Resistor Technique for Process and Temperature Compensations of Low Power Ring Oscillators," 2019 IEEE 10th Latin American Symposium on Circuits & Systems (LASCAS), 2019.

[6] J. Zangpo, R. Póvoa, J. Guilherme and N. Horta, "An Integrated LC Oscillator with Self Compensation for Frequency Drift and PVT Corners Variations," 2018 25th IEEE International Conference on Electronics, Circuits and Systems (ICECS), 2018.

[7] V. Jahagirdar and R. Hirur, "CMOS current reference with temperature compensation," 2015 IEEE International Conference on Electrical, Computer and Communication Technologies (ICECCT), 2015.

[8] Y. Moursy, R. Iskander and M. Louërat, "Automated triangular wave generator design with process corners compensation," 2015 IEEE 20th International Mixed-Signals Testing Workshop (IMSTW), 2015.

[9] W. L. Jiang et al., "Monolithic Integration of a 5-MHz GaN Half-Bridge in a 200-V GaN-on-SOI Process: Programmable dv/dt Control and Floating High-Voltage Level-Shifter," 2021 IEEE Applied Power Electronics Conference and Exposition (APEC), 2021, pp. 728-734.

[10] S. K. Murray et al., "Transient Overvoltage Detection Technique for GaN HEMTs Integrated in a 200-V GaN-on-SOI Process," 2022 IEEE Applied Power Electronics Conference and Exposition (APEC), 2022, pp. 1400-1405.

[11] D. Bisi et al., "Short-Circuit Protection for GaN Power Devices with Integrated Current Limiter and Commercial Gate Driver," 2022 IEEE Applied Power Electronics Conference and Exposition (APEC), 2022, pp. 181-185.

[12] P. S. Niklaus, D. Bortis and J. W. Kolar, "High-Bandwidth High-CMRR Current Measurement for a 4.8 MHz Multi-Level GaN Inverter AC Power Source," 2021 IEEE Applied Power Electronics Conference and Exposition (APEC), 2021, pp. 200-207.

[13] Walt Jung. "Sources 101: Audio Current Regulator Tests for High Performance" AudioXpress magazine. 2007, pp 1-10. (https://audioxpress.com/assets/upload/files/ Sources_101_P1.pdf)




Authorized licensed use limited to: Tsinghua University. Downloaded on February 12,2026 at 02:45:44 UTC from IEEE Xplore. Restrictions apply.